

THE
F L O R I S T
AND
HORTICULTURAL JOURNAL,

A MONTHLY MAGAZINE OF

**Horticulture, Agriculture, Botany, Agricultural
Chemistry, Entomology, &c.**

H. C. HANSON,
EDITOR,

WITH THE FOLLOWING CONTRIBUTORS :

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VOLUME II.

PHILADELPHIA:

1853.

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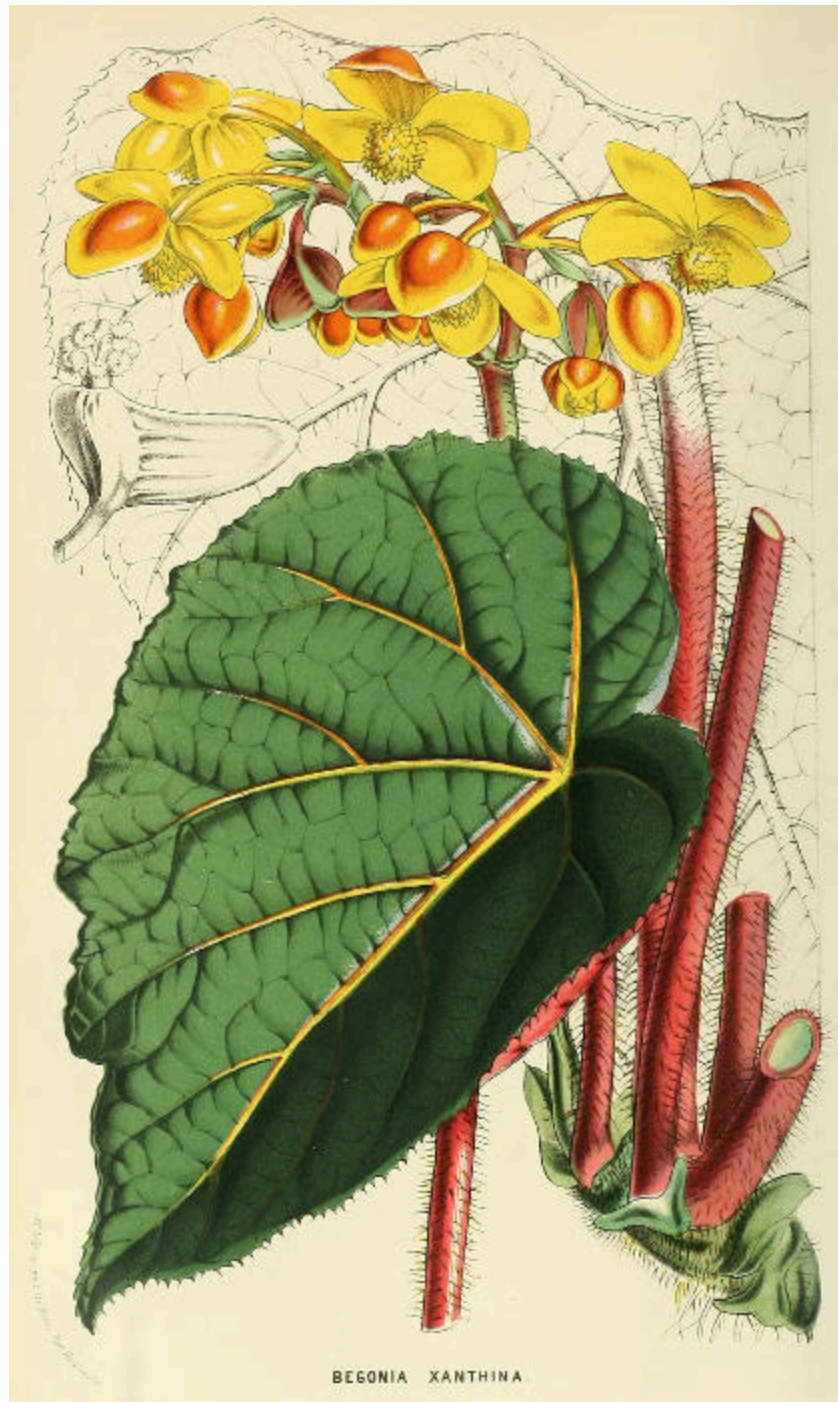
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BEGONIA XANTHINA.

THE FLORIST AND HORTICULTURAL JOURNAL.

Vol. II.]

Philadelphia, July, 1853.

[No. 7.

BEGONIA XANTHINA.

YELLOW-FLOWERED BEGONIA.

Begoniaceæ.—Monœcia-Polandria.

CHARAC. GENER.—Flores monoici. MASC. Perigonii tetraphylli foliolis subrotundis, 4 exterioribus majoribus. Stamina plurima; filamentis brevissimis liberis v. basi-connatis, antheris extrorsis bilocularibus, loculis linearibus discretis, connectivi continui obtusi margini adnatis longitudinaliter dehiscentibus. FEM. Perigonii tubo triptero cum ovario connate, limbi superi 4-9-partiti persistentis lobis pluriseriatim imbricatis. Ovarium interum triloculare. Ovula in placentis e loculorum angulo centrali bilamellatis plurima anatropa. Styli 3 bifidi stigmatibus crassis flexuosis v. capitatis. Capsula membranaceo-trilobata trilocularis loculicide trivalvis. Semina plurima minima striata. Embryo in axi albuminis carnosus orthotropus.

Herbæ in Asia et America tropica indigenæ, foliis alternis petiolatis integris v. palmatilobis basi saepe cordatis inæquilateris integerrimis dentatis v. mucronatoserratis stipulis lateralibus membranaceis deciduis, cymis axillaribus pedunculatis dichotomis floribus albis roseis v. rubicundis.

CHARACT. SPECIF.—B. acaulis, rhizomate brevi crasso subter radicante, foliis amplis oblique cordato-ovatis brevi acuminatis sinuatis denticulatis subter discoloribus (rubris) petiolis aggregatis crassis folium subaequantibus rubris stipulatis crinitis, setis patentibus interioribus reflexis, scapo petiolis duplo longiore, floribus nutantibus corymbosis flavis, masculis tetrasepalis, sepalis 5 oblongo-cuneatis unico majore rotundato magis concavo, foeminis triplo minoribus hexasepalis, sepalis æqualibus ovali-rotundatis, fructus alis duabus brevibus unica horizontaliter elongata striata. HOOK.

Begonia xanthina, HOOK, Bot. Mag. t. 4683.

Although many different species of this valuable genus have been discovered up to this time, we have had only those with white or red flowers. *B. cinnabarina* with its orange red flowers was an approach to what we now figure—the *Begonia xanthina*. But even the yellow of this is shaded with the red which prevails in a greater or less degree in the flowers or leaves, and stalks of all the genus.

This species, which flowered in July, 1852, in the collection of Mr. Nuttall, at Rainhill, Lincolnshire, was received by him in 1850, from the Bootan Himalayas, having been sent thence by his nephew, Mr. Booth.

It may be some time before this beautiful plant is imported into this country; but we know that our enterprising nurserymen and amateurs will obtain it as soon as it finds its way into the hands of the trade in Europe.

HISTORY AND CULTIVATION.

Few plants have a greater claim on the American plant grower than the *Begonia*. It has been too much the habit to sigh after, and bewail the want of “Chiswick Heaths,” and other things which *do not* do well in America, to the manifest neglect of many beautiful things which *do*. It is time we had ceased to be the mere copyists of English horticulture. We have so rapidly advanced, that we should aim at an independence that can be achieved; and, as in government so in gardening, take our place as one of the horticultural “nations of the earth.” We have been a “colony of Chiswick and Edinboro,” “Paris and Ghent” hitherto; we have experienced on every occasion slights and neglects; whatever we do is passed over in silence, and whatever we discover remains unnoticed or is scorned. These are some of our grievances. All our horticultural papers have taken up the subject in turn, and pressed our claims on English journalists; but how have they been met? A private letter on the success of one individual plant has been published in one magazine; and two hybrid Peonys have been named in Belgium in honor of Americans. Perhaps once a year a short extract in the *Revue horticole* on *Forsythia viridissima* from the *Horticulturist*; or, a notice in the *Gardener’s Chronicle* of how to preserve Tomatoes from *Hovey’s Magazine*. We must have done with whining and complaining about these things. Let us strike out new courses for ourselves. We may never hope to excel them in Heaths, Pansies, Calceolarias, or many other things, as a general rule, nor is it desirable we should. Let them boast of their excellence; we will raise another standard.

The BEGONIA is peculiarly adapted to become such a plant as I have described. Requiring in England a moist and very artificial atmosphere, it does not make any very great progress in popular estimation. Here it thrives with very common care; all doing in a greenhouse 9 months in the year; and many doing well in the open air, if in a somewhat shaded situation. They are for the most part natives of Brazil or Mexico.

To cultivate them successfully we must divide them into two classes:—the *tuberous* rooted, and the *shrubby*. Each of these will require separate treatment. The following kinds are amongst the best in cultivation, either here, or in English gardens, from whence they can be easily procured.

TUBEROUS ROOTED.

1. *B. discolor* or *Evansiana*, native of China, with pinkish white flowers, may be had in bloom from May to October.
2. *cinnabarina*, from Bolivia, deep pink or cinnabar, from June to October.
3. *Martiana*, from Brazil, deep pinky rose, from July to October.
4. *diptera*, Cape of Good Hope, whitish, June to August.
5. *Barkeri*, Mexico, dull white, February to December.

These require to be kept rather dry and cool in the winter season. No. 5. may be had in flower all the year, by having several pots, and keeping them dry at different periods. Early in spring the tubers may be potted in 6 in. pots, in a soil composed of well-decayed leaf mould, loam, and sand. They require little water till the leaves appear, when they will take an abundance. They may be forwarded in a little heat, but will do pretty well if allowed to come along with the season. They are easily propagated from their tubers, by cuttings, or by seeds.

SHRUBBY OR FIBROUS ROOTED.

1. *B. nitida*, native of Jamaica, with pinkish white flowers, and large, thick, shining leaves. May be had in bloom all the year, and made into handsome specimens.
2. *B. spathulata*, another West Indian, with a very erect habit of growth. The leaves are folded in like spoons, and the small white flowers appear at the ends of the young growth. It flowers from August to December, and is but an average kind.
3. *B. odorata*, a Brazilian kind allied to the last, but has a fine foliage, resembling *nitida*, the flowers came out like the last from August to December, are much larger and sweet scented.
4. *B. hirtella*, a Brazilian species, with a rather starved looking habit of growth, but an abundance of small, pinky white flowers, appearing from June to December.

5. *B. ulmifolia*, a South American, with elm like leaves, but of no great beauty of flower. Only desirable to form a collection.

6. *B. argyrostigma*, a Brazilian species, with curiously spotted leaves, which is its chief attraction. The flowers appear from June to December. It is a very strong grower—occupies much room.

7. *B. hydrocotylifolia*, a Brazilian pretty species. It is herbaceous, or has its leaves from a rhizoma creeping on the surface. Its pink flowers, borne on scapes about a foot high, appear from February to May.

8. *B. parvifolia*, a native of the Cape of Good Hope. It has very small leaves, grows about 2 feet high, and is always in flower. A white flowering and desirable kind.

9. *B. albo-coccinea*, a West Indian. Its oblique leaves are almost round, are very large and thick, and of a deep red beneath. The flowers appear from February to May; white on the inside, and scarlet on the back. The best of the rhizoma producing kinds.

10. *B. nomonyma*, a Brazilian species, in the way of *B. manicata*, with small white flowers, continuing from July to November.

11. *B. castaneæfolia*. The specimens that I have seen growing are so like *B. ulmifolia*, that I have either not seen the true species or they are both the same.

12. *B. Fischeri*, a rather scarce Brazilian species, but one of the most desirable, approaching *B. incarnata*. In the spring months it is covered with its numerous pink or white flowers.

13. *B. incarnata*, a South American, that should be No. 1 in all collections. Its pink flowers may be had nearly all the year.

14. *B. coccinea*, another first rate Brazilian. A strong grower, covered with spikes of scarlet flowers from April to July.

15. *B. manicata*, from Brazil. A fine species when well grown, though the individual flowers are indifferent. The leaves are distinguished by a production resembling the claws of a mole on their under surface. Flowers in winter and spring.

16. *B. fuchsoides*, a well known species from New Grenada, with fuchsia like foliage, and scarlet flowers, appearing from June to September, a strong grower and fine kind.

The species of this division are readily propagated from cuttings of the half ripened wood, put in sand, and plunged in a slight bottom heat. They are liable to damp off if kept too close. Indeed if they are in a situation somewhat shaded they will do better without the usual accompaniment of a bell glass. They will grow well in a soil of sandy loam and leaf mould. They should never be grown in very large pots; or, in the language of gardeners, should be always under potted. When growing fast they take a good supply of moisture; love a moist atmosphere and frequent applications of the syringe, and may be placed in the full light. If a moist atmosphere cannot be maintained, they will do better in a shaded part of the green house. The chief thing to guard against, is their damping off; small, well drained pots are the securities. When they are not growing they will live and do better with very little water. They are easily raised from seed, sown on the surface of the soil in pans, and placed in a warm shaded place, with the only attention of never being allowed to get dry. Some fine varieties, I believe, have been lately raised by hybridizing in Europe. It opens a fine field.

I think the remark of the Calendar writer in the last No. worthy of repetition; that the Horticultural Society would do well to encourage the growth of the Begonia by a premium.

A PHILADELPHIA GARDENER.

THE AURICULA.

Dear Sir:

Allow me to draw the attention of your readers to that much neglected but truly desirable class of flowers the Auricula. There appears to be a general idea prevailing amongst many persons that considerable difficulty is experienced in cultivating this lovely gem in our climate. This notion is certainly erroneous, and the object of the following remark is to show how this may be accomplished. The Auricula, (*Primula auricula*,) is one of the very numerous species of Primrose, and no mean species either. Linnaeus claims this genus for Pentandria monogynia, and it forms the type of the natural order Primulaceæ. Our present subject is indigenous to the alpine districts of the European continent, Syria, and occasionally is found in the same situations in Britain, though rare. Its name Auricula has been applied from the supposed resemblance of the form of the leaves, to the ear of an animal, and hence the vulgar cognomen “Bear’s Ears,” a name somewhat revolting to the taste of some of our delicate and sensitive belles, but quite in character with the quaint and uncouth, yet intelligent and enthusiastic class of individuals, with whom it originated. In a wild state, the colours are yellow, purple, and variegated, and I am inclined to think, in opposition to the acknowledgement of some botanist, that two or three which are considered as distinct species, are nothing more than varieties of this, if so we may include white also. Handsome as the different varieties of this flower appear naturally, the claim to beauty has been so much enhanced, by the perseverance and enthusiasm of the florist’s fostering care, for the last three hundred years, that at length, it has assumed a perfect symmetry of outline and marking, which renders it truly a gem. Were it only for the peculiarly rich odor of the flowers, it deserves a place in every garden, but when we combine this with the evergreen and neat habit of the plant, and the exquisite beauty of the flower, it seems strange as the cultivation is so easy, that it is not more generally seen, even what is grown are mostly varieties of no pretension to perfection, but simply a step or two removed from the natural state.

The Auricula is divided by florists into four classes, viz, green edged, gray edged, white edged, and selfs, the edged classes being mostly esteemed as exhibition flowers, although the selfs are tolerated and encouraged. There are also, several double varieties, but these are not considered equally valuable, yet they are well worthy of attention.

The following criteria constitute what is considered to be the main points of excellence in a prize Auricula. The stem should be strong, erect, and high enough to raise the truss of flowers above the foliage. The individual footstalk, sufficiently strong to support the flower, and of a proportional length to the number of pips, so that they may not crowd each other, and which should not be less than seven in number, that the truss may be close and compact, and form somewhat a half globe. The tube containing the anthers, the eye, and the exterior circle, ought to be well proportioned, which will be the case if the diameter of the tube be one part, the eye three, and the whole pip six or seven. If edged the margin should be about equal with the next inner circle. The edges ought to be smooth, having no serrature, so as to appear starry, and the limb or upper surface, flat and even. The nearer the outline approaches to a perfect circle the better, although the very best flowers do not quite come up to this point as yet. Whatever the colours, they should be clear, bold, and distinct, and divided in a perfect circle, or the dark markings form a circle next the eye, and extend out towards the outer rim on each lobe, so as to form so many half circles. In the selfs the colours ought to be uniform, bright and solid, or shaded off towards the outer margin distinctly and clear.

Around the cultivation of nature's greatest beauties there is generally a halo of mystery thrown. When any thing of this kind becomes recognized as a general favorite, speculative ideas, and vague theories have each a portion of precedence, and as some of these peculiarities happen to succeed, they are lauded, made public, and the tyro catches up the most ridiculous notions. Our present subject has not entirely escaped from this general contagion, for if we refer to some old and long established growers of the Auricula, we find that one thinks that the rotted down roots of the Willow, is the only matrix in which it will approach perfection; others again suppose, that nothing is so suitable as rushes decomposed into mould, and a portion of the same material cut short in a fresh state, and strewed over the drainage; while some will not believe in any fertilizer, but blood, mixed up

with maiden earth, and laid together for a season. Now all these materials are well enough in their place for other things besides the Auricula, but to say that they are absolutely necessary, and that nothing else will produce the same quality, partakes of a superstitious and retrograde movement, and belongs to the old school. The fact is, all that is required, is a tolerably rich and cool base, for if too poor the plant will be weak and the flowers correspondingly so, and if too rich the trusses are apt to become monstrosities, having irregular shaped flowers, and too crowded to form a handsome bunch, with confused colours, and undefined markings. In these remarks, I would not wish to detract from the honor due to the old floral veterans, to whose zeal we owe a debt of gratitude, whose exertions have produced many of the very best flowers, and without which we should have been minus of many classes of the greatest beauty. Still we must move ahead we live in an age of progress, and if we cannot accomplish greater individual perfection, we can certainly attain to the same by more simple means. To grow the Auricula, plant in a tolerably rich soil, a suitable compost may be formed, by mixing two thirds fresh loam from a pasture, and one third cow dung, well rotted, and laid in a heap a few months before being used. Fresh dung injures the colours, and causes the plant to grow too rank and deficient in substance, when thin and papery flowers are the consequence.

The Auricula may be grown in pots, or planted out into frames, or the open ground. If it is desirable to have a show of this flower in the greenhouse or parlour window; about the middle of August, prepare a quantity of six inch pots, place in the bottom of each, about an inch of broken crocks, or what is better the same depth of lumps of charcoal, over this put a layer of moss or fibry turf, to prevent the soil from falling down amongst the drainage, next fill in a portion of the above mentioned compost, and place into the centre of each pot a good and strong single crown. Before planting examine the roots and cut away any decayed portions, spread them out carefully, and fill up to the rim with soil, press it a little tight about the crown, and give the pot a smart rap or two upon the potting bench, which will settle all and leave a little space for future watering, do not plant too deep, but let the collar of the plant be somewhat elevated, which will prevent damping off, after potting give a good watering, but do not repeat this afterwards further, than to keep the plant in health, for the Auricula is impatient of too much moisture, and most

particularly so while at rest in the winter. In order to avoid continued saturation, the pots should be placed in a position where the water can pass away freely, and likewise shaded from the sun's rays; if plunged in a bed of sand or porous ashes, so much the better, and if worms should get into the pots, give it a soaking when dry with lime water, using the clear liquid; here they may remain till the approach of severe weather, when the pots ought to be washed and the plants carefully looked over (all decayed leaves should be removed clean away,) when they may be taken into the greenhouse and placed near the glass, freely exposed to the air and light; water carefully and rather sparingly at first, afterwards gradually increase it, (but at no time unduly,) as they advance towards blooming; do not give much artificial heat, or the flowers will be rendered weak or abortive; a Camellia or Geranium house is quite hot enough, in which a temperature of 45 at night ought to be kept.

Where there is not the luxury of a greenhouse, the same object may be accomplished by a common garden frame, (in fact a much better display to my taste) and with much less trouble and expense considering the great number of plants that may grown. In this case, during the latter part of August, or early in September, place one or more frames, according to the number of plants, in a situation freely exposed to the sun, and upon a well drained bottom, dig up the base so that the water may pass freely through, and fill in with about eight inches of the before mentioned compost, divide the plants into single crowns, the same as recommended for pots, and plant about six inches apart; if dry weather give a good watering, and keep off the glass till frosty weather sets in, when the sashes may be put on, and air admitted freely through the day. In case of rain the lights may be tilted up at the back, which will keep the plants dry and prevent any danger from rot. Open early in the morning, and keep all at rest till towards the middle of January, when the lights may be closed somewhat earlier in the evening, and the sun's heat allowed more freely: cover with straw mats or other material, to keep out the frost, and line round the sides with rough litter or clean straw for the same purpose. When the flower trusses commence to push, water may be more freely given, but not overhead, unless in case of a genial showers, which seldom occurs at this time, and after the pips begin to expand, do not let even the rain in upon them, as much of the fine powdery substance, which constitutes a great portion of their beauty, would be thereby washed off, but apply water more freely between the rows, admit

air sufficient, but avoid beating winds. By following these directions, and having a good stock of plants, the lover of flowers may be rewarded through the months of March and April, without the aid of a greenhouse, with one of the loveliest sights, and most enchanting scents in Flora's kingdom. So far we have spoken of the little favorite as a nursling and treasured up pet, but like many other of God's blessings, it is not so very mindful of man's fostering care. There are many of our city yards in which there is a shaded and sheltered spot, in all such there is ample accommodation for this plant; use a fertile and well drained soil, screened from the burning orb of summer, and the thawing gleams of winter, and we can be recompensed by a sight of its beauty, and refreshed by its delicious odor, without any other protection, although of course if protected and taken care of under glass, we have greater perfection, and shall be no losers for our extra trouble.

After blooming is fairly over, choose a piece of ground sheltered from the midday sun, if moist but yet drained, so much the better, plant out in rows a foot apart, and six inches in the row, place a little deeper than before, which will cause them to push out side roots. If dry give a good watering when planted and an occasional repetition, in case of drought through the summer; no further care is required, unless Red Spider, or Aphides should attack, when a good syringing with soap-suds, will speedily eradicate them.

The following list, containing twelve of the best in each class, is from the authority of Mr. John Slater, one of the "Lancashire Heroes," and a sure guide, a man whose enthusiasm knows no bounds, and whose honesty is equal to his perseverance.

GREEN EDGED.

Litton's Emperor, Leed's Colonel Taylor, Booth's Freedom, Pollitt's Highland Laddie, Ashton's Prince of Wales, Pollitt's Standard of England, Yates' Morris Green Hero, Oliver's Lovely Ann, Barlow's King, Howard's Nelson, Moore's Jubilee, Page's Champion.

GREY EDGED.

Cheetham's Lancashire Hero, Syke's Complete, Kenyon's Ringleader, Grimes' Privateer, Fletcher's Ne Plus Ultra, Fletcher's Mary Ann, Waterhouse's Conqueror of Europe, Rider's Waterloo, Kent's Queen Victoria, Taylor's Plough Boy, Beeston's Fair Flora, Ashworth's Newton Hero.

WHITE EDGED.

Taylor's Favorite, Lee's Venus, Ashworth's Regular, Taylor's Incomparable, Taylor's Glory, Wood's Delight, Catharina, Popplewell's Conqueror, Kenyon's Lord Chancellor, Hepworth's True Briton, Cheetham's Countess of Wilton, Pott's Regulator.

SELFS.

Netherwood's Othello, Kay's Jupiter, Blegg's Blue Bonnet, Berry's Lord Primate, Berry's Lord Lee, Grimes' Flora's Flag, Redmayne's Metropolitan, Schole's Ned Lud, Whittaker's True Blue, Bradshaw's Tidy, Barker's Nonsuch, Gorton's Stadtholder.

Yours Most Respectfully,

WILLIAM CHORLTON,
New Brighton, Staten Island.

June 27, 1853.

GARDENS OF PAEONIA MOUTAN IN CHINA.

BY R. FORTUNE.

These gardens are situated near the village of Fa-who, about 5 or 6 miles west from Shanghai, in the midst of a vast country of cotton plantations. On the way I met a large number of *coolies* each carry two baskets of Moutan-paeonies in full flower; they went to sell them in market. Arrived at the gardens I found there a number of these plants in flower and of a remarkable beauty. The purple and lilac species especially attracted the sight: among others a very pretty kind apparently distinct, with finely cut leaves and flowers of a velvety purple, like the Tuscan rose of our gardens. The Chinese call it Moutan Peony (black) and I believe it to be the same as that called *atro-sanguinea* by Dr. Lindley, in the journal of the Horticultural Society of London. Another species called “tse” or purple, had double flowers of a remarkable size; it is probably this variety which they say has a thousand petals, and did not exist but in the garden of the emperor. The third kind is called “lan” (blue;) it is a lilac variety; its flowers are of the color of *Wistaria Sinensis*. There are besides other kinds of purple differently shaded, very distinct from the preceding and equally beautiful.

The double whites are also numerous and very remarkable. The largest among them has been called *P. globosa* by Dr. Lindley, but there are 4 or 5 others which approach very near to this. Some have a light shade of lilac, which gives tone to their color. The best is that called “wang” (yellow) by the Chinese; this variety, of a straw color, is very beautiful, but notwithstanding inferior to some others.

The red peonies (Hong) are equally numerous. What is strange, is, that the kinds common at Canton and in England, are very rare here. These gardens contain about a half-dozen of new varieties of red peonies; among others, that called “Van-yang-hong” by the Chinese, is the most beautiful plant which I have ever seen. Its flowers are of a bright and pure red, entirely different from all the others perfectly double; each of them is about ten inches in diameter. In all I counted nearly thirty distinct varieties in these gardens.

The greater part of these beautiful varieties of *Paeonia Moutan* are unknown at Canton. This may seem strange in a country where flowers are so generally sought for; but the Chinese are in everything such conservatives that a slight acquaintance with their customs will suffice to explain this apparent anomaly. The gardens of Canton derive their supplies of *P. Moutan* from a district situated more in the west than Shanghai. From time immemorial the same gardens have furnished these flowers; they arrive always by the same road, and at the same time of the year. It seems that Shanghai until the end of the last war, has never been in communication with Canton, at least in what concerns flowers, consequently these beautiful varieties of Peony could not obtain a route towards the south and thence to Europe.

The establishments where they cultivate exclusively the *Paeonia Moutan* are numerous, but very small. They have the appearance of our cottage gardens, and are cared for in the same way, that is, by all the members of the family: the women take as much part in it as the men: they are very avaricious and love money extremely. When they have been consulted, I always had to pay dearer for the acquisitions of plants which I made. The soil of these gardens is rich and well-manured; this latter circumstance renders it less compact than the soil where they cultivate cotton.

The propagation and management of the *Paeonia Moutan* seems to have been perfectly understood by the Chinese at Shanghai, much better than with us. Our horticulturists complain without ceasing of the difficulty of multiplying this beautiful plant, and this makes the price remain high. Here is the method which the Chinese pursue, and which our growers may try.

At the commencement of October, they collect in the sheds and out-houses a great quantity of the roots of a certain herbaceous Peony, roots which must serve for subjects of *Paeonia Moutan*. The bundle of tubercles which forms the root of a herbaceous peony is divided, and each little root, in shape of a finger, becomes a subject upon which they graft the *P. Moutan*. Having placed a great number of these tubercles upon the potting table, they bring the grafts of the plants which they wish to multiply. Each graft is but an inch and a half to two inches long; it is the extremity of a shoot made during the summer just finished. The under part of it is cut into a wedge and inserted on the top of the finger shaped tubercle of which we spoke. The graft is covered by a ligature or by clay and the operation is complete. A

great number of grafts being thus made, they are carried to the garden and planted in rows distant about a foot and a half; the same space is left between plants in the same row. In planting, the head of the graft only remains above the soil; the part where the graft unites with the tubercle is always buried. Kaempfer says that the Chinese multiply the Moutan Peony by bud-grafting; this is an error, this process is never practiced by them, they do not even understand it. The author has been led into the error by the smallness of the bud which they use, this having but one eye at its end.

Each autumn sees thousands of plants grafted in this way. The little empty space which may be seen in the rows proves the goodness of the method; in fact a graft rarely fails to grow. At the end of about fifteen days the union of the root and the graft is complete: in the following spring the plants are beautiful and vigorous. They generally flower the first spring or at the latest in the second; it is then that they are taken up and carried to be sold in the market in the way I have described. The plant which has but one stalk and one flower has more value in the eyes of the cultivator at Shanghae than a stronger one; it is sold more easily, produces a very large flower, and presents no difficulty neither for lifting nor for transportation. Thanks to this circumstance I could always procure strong plants more advantageously than small ones.

In the gardens of the mandarins can often be met Peonies of a considerable size. Near Shanghae there is one which annually produces 3 to 400 flowers. The proprietor takes as much care of it as could the most enthusiastic amateur of his tulips. During its flowering, it was protected from the burning rays of the sun by a tent of canvass; in front was a seat where the visitor enjoyed in full the sight of these magnificent flowers. Every day, for several hours, the old man installed himself there, and while pipes and bowls of tea succeeded each other he looked lovingly upon his favorite “Moutan wha.” It was certainly a noble plant, well worthy of the admiration of the old amateur, to whom I wish the pleasure of sitting a long time yet under his tent, to enjoy so beautiful a sight.

Translated from the Flore des Serres.

CHINESE PRIMROSE.

Winter flowering plants are not too plentiful, at least such as will bloom in the cool temperature of a greenhouse; we have many fine winter flowering exotics, but they either require a hothouse temperature to bloom freely, or are more difficult to manage during summer; we are indebted to the Celestials for many of our most valuable winter flowers, Azaleas, Camellias and among others that beautiful little gem the Chinese primrose. They are all of hardy and robust habit, and are thus within the reach of all, so far as cultivation is concerned, a recommendation of much importance to those amateurs who cannot devote much time to their collection. It has often occurred to me when visiting amateurs' greenhouses, that they would derive much more pleasure from their labors were they more select in the choice of their plants, choosing those only of known hardy and free blooming qualities, instead of aiming at a varied assortment, often expensive, and rarely satisfactory. I was more deeply impressed with this fact during a visit to an enthusiastic amateur last winter. On entering his greenhouse I was struck with the gay and cheerful appearance it presented, and it was only when about leaving that I observed the only plants in flower were a few Camellias, two or three Cinnerarias, and a quantity of Primulas of various colors, but these were so well arranged and set off with a profusion of healthy foliage, and the Primroses showing such fine rounded trusses, that I am confident not one person in ten would have noticed this simplicity of its production.

Although this plant is very hardy, I am aware that many amateurs are not so successful with it as they would wish. This, I presume, arises principally from its liability to decay if kept too damp at root, either from using pots of a large size, or supplying water too freely. Four and six inch pots will be found of ample size, not that a plant will not grow in a pot three feet in diameter as well as it would in one inches. The only difference would be that, in the former, the plant would probably require water once a month and the latter daily.

To keep up a succession of flowers for four or five months, seed should be put in at intervals of five or six weeks, from June to September, soil of a sandy nature should be prepared, and the seeds very slightly covered. The front shelf in the greenhouse will be a suitable place for them while growing, they should be encouraged by transplanting into a shallow box as soon as the first true leaf makes its appearance. If thinly planted (not closer than three inches) they can remain until sufficiently strong to be placed at once into flowering pots. These must be well drained by placing a handful of broken material in each pot. If attention is paid to this, there will be little fear of damp, if ordinary care is given in watering. Two small stakes should be inserted one each side of the plant, crossing other at top, no tying will be necessary, this will prevent them from toppling over and breaking off at the neck where they are very weak. A few of the latest sowing should be set apart for seedling. A little care and attention should be given to their selection, choosing those with large, well colored fringed flowers. Sometimes they incline to be semi-double; these should have the preference. About the middle of May they should be set out in the sun, watered regularly, and if they receive an occasional application of manure water the value of the seed will be enhanced. Under these conditions an abundance of seed will be secured. It is not worth trouble to preserve old plants; young ones flowering with much more vigour; any particularly fine variety, however, may be turned out of the pot after it has done blooming, some of the old soil removed and replaced in fresh compost. The double varieties are much superior to the single ones. They are increased by dividing the plant into cuttings. To propagate them requires a cool dry situation, partially shaded, paying great attention in watering, they must be kept rather dry until they begin to show symptoms of growth; otherwise they are certain to decay. It is probably on this account that they are not so extensively cultivated as they deserve to be.

DELTA.

ON THE DOUBLE CHINESE PRIMROSE.

To the Editor of the Florist, Sir: As you invite communications from practical gardeners, I offer a few remarks on the above; as I think it a much rarer plant than it ought to be—few plants excel it in simple beauty, and by proper management it may be had in bloom every month in the year. Being double, it can only be propagated by cuttings, which may be taken off at any time, and readily struck in sand under a bell glass. In about three weeks they are sufficiently rooted to pot into single pots. I find them to thrive best in a compost of rough leaf mould and sandy loam, in well drained pots. I repot them as often as they become well rooted; and pinch off every flower stalk that appears till the plant is six months old at least. I then let it flower on, and as each flower truss shows symptoms of withering, pluck it out. By this course it will continue to flower a whole year. They frequently die out from exhaustion at this time, and the only way I find to prevent this, is to divide the plant into off-sets, and so start with them as with new plants. They always do well with me by this treatment. The double white is the commonest and most easy to be procured; but I have observed in a report of the Pennsylvania Society, that the purple exists also in your neighborhood.

If you think the above worthy of insertion in the Florist, I should be happy to send you some others when I have leisure.

Yours, &c., HORTUS.

NOTES ON ROSE CULTURE, BY A LOVER OF ROSES. (Continued.)

HYBRID PERPETUAL OR REMONTANT ROSES.—This division of the Rose was introduced to us about 15 or 16 years ago and according to the opinions of good judges it originated between the perpetual Rose Du Roi and the Bourbon Rose; they have been till very recently nearly all of a crimson or dark red color. Florists and Rose-growers generally esteem these the finest of all Roses, to which I would coincide, provided, they were constant monthly bloomers, which they are not, giving only two or perhaps faintly a third bloom with a meagre flower in the autumnal months. They richly deserve special attention for their fine bold flowers, rich foliage, and luxuriant shoots when on their own roots; or if budded they make very admired standard Rose trees. But oh! what nicety of eye, how critical the judgment to detect the difference of one half of them. For my own pleasure and gratification I would not give a groschen for $\frac{3}{4}$ ths of them; but we are so led off our guard by “have you got so and so,” we blush to own up, and buy nearly all, good, bad, and indifferent; but pray who would grow Doctor Marx, Dr. Margolin, Madam Laffay, and nearly 50 others; all red, rosy red, dark red, light crimson, and such transpositions, meaning nearly the same in color and character? So be it. I will not detail before your thousands of readers such equivocal terms, but give off hand and full in my eye, a score that will please the most fastidious.

Auguste Mie, clear waxy rose, cupped petals.

Baronne Hallez, dark purple crimson, fine form.

Baronne Prevost, rose color, very large.

Caroline de Sansal, large, rosy blush, pink centre, of remarkably strong growth.

Duchesse de Montpensier, pale satiny rose, very fragrant.

Giant of the Battle, nearly bright scarlet. Is this not the rose of the group? Everyone of your readers that wishes a rose bush that is perfectly hardy,

bright in color, and constant in bloom may chose this subject.

Julie de Krudner, a new rose, nearly white, very double, and agreeably fragrant.

La Reine. I hesitate about this though a most superb flower; it has a paucity of bloom not altogether to my fancy; the color too is undecided being a rosy lilac, but extremely large and luxuriant in growth.

Lion des Combats, reddish crimson, shaded with scarlet, large, full, and fragrant.

Louise Le Clerc, delicate blush, perfect form, very profuse and fragrant.

Madam Laffay, purplish lilac, inclining to crimson, very full regular flower, plant of strong growth.

Madam Rives, quite a new variety, of a pale flesh or silvery blush color, of great substance, very distinct and perfectly formed.

Marquis Boccella, pale blush, dwarf habit, a constant bloomer, very distinct.

Pius 9th, crimson purple, very large, full and perfect, a profuse bloomer of strong growth.

Patenotte, pale rose, very exquisite form, fragrant and profuse, quite a new variety.

Queen Victoria, (Paul's) blush white, shaded with pink or peach blossom color, large, very full, and distinct from any other variety, quite new, very desirable.

Robin Hood, bright carmine, beautifully globular, and a rampant grower.

William Griffith, satiny rose, distinct color, a large bold petaled flower, quite new, and will rank amongst the best.

William Jesse, a favorite variety of a rosy crimson color, edged with lilac, very large, full, and constant.

Yolande d'Arragon, pale rose, producing its flowers in clusters the whole season, a strong grower. Sydonie, a newer variety, is very like this, and I am doubtful if it is any improvement upon it, unless to a very critical observer.

I have not seen a clear good white as yet in this group. Florists and catalogues tell us of *Blanche Vibert*—white it is, but a sickly looking flower and plant, at the best. *Blanche Portemer* has nothing more attractive about it. A good white such as the old *Madam Hardy* is much wanted, it would be a treasure. There are over 100 of these Hybrid Perpetual varieties cultivated, one half of them are nearly facsimiles of each other. They all require more stimulation in their culture than any other roses; if a strong luxuriant growth is not produced a very meagre show will follow; dark rich green foliage and strong shoots will produce a corresponding bloom—but yellow foliage and weak growth produces similar sickly blooms, deficient in color, size, and fragrance. Also, observe that taking off one half of the buds in May and June, is beneficial to a continuance of bloom during the season; in dry weather they should be liberally supplied with rich water or soap suds, or if the ground has been abundantly manured, water of any description will suit; observing that one copious supply once a week is better than a small portion every day.

NOISETTE ROSES are those clustering sorts that bloom from early in the season till destroyed by frost; their flowers are generally small, though there are now some of them nearly as large as any others, but the size takes from their profusion. The growth of the family is of all grades, from 1 foot to 20 feet in a season; unfortunately for us, however, the finer kinds are rather tender, being nearly or entirely killed in our severe winters. Those tender kinds of greatest splendor are *Lamarque*, *Cloth of Gold*, *Solfatare*, *La Victorieuse*. These are hybridised with the Tea Rose, making them more tender; but for milder climates than Philadelphia the splendor of them equals any description. A friend writes me from Texas, that *Cloth of Gold* reaches the 3d story, and is in bloom 9 months of the year, and its odor equal to the magnolia. In pruning Noisette Roses, thin out the old wood, and shorten the wood of the previous season. The young wood should always be allowed to have its full growth, as it is most frequently terminated with a cluster of flowers; the sorts without encroaching upon their beauties might be curtailed to one full dozen, which would comprise as follows:

Aimie Vibert, pure white, very compact, a dwarf grower.

Cloth of Gold, a very luxuriant grower, requiring dry rich soil; flowers quite large, frequently 5 inches in diameter, opening a bright lemon yellow,

and fading to a pale straw color; should be protected by covering in winter.

Desprez or Jaune Desprez, flowers buff color shaded with pink, in profuse clusters, very fragrant, plant vigorous.

Fellenberg, color pink, red, or crimson as the season may be; a constant bloomer—does very well for a piazza or pillar, quite hardy.

Lafayette, this old fine pink rose must not be neglected; there are few or none to equal it in color and profusion; a moderate grower, flowers prettily cupped and quite profuse.

Odorata, pure white, very sweet, an abundant bloomer and perfectly hardy; stood the winter of '51, without any protection; of moderate growth.

Ophirie, color orange, orange and pink, pale yellow or bright yellow, all according to the weather and season; blooms abundantly, especially in the autumnal months, flowers very double, plant hardy and growing freely; quite distinct, foliage of a pleasing lively green.

Philippart, dark pink, must be well established before its character is fully ascertained; blooms in large clusters; exceedingly hardy and of strong growth.

Pourpre de Tyre, not easily to decide whether this variety is a Bourbon or Noisette; but florists place it where I now do. The flowers of medium size, dark red or crimson color; plant of moderate growth.

Triomphe de la Duchere, one of the very best Noisette roses, flowers abundant, large, full, of a rosy pink color and fragrant; plant strong, but not rampant.

Vittelina, pure white, fine form, buds shaded with pink, very double, medium size, growth moderate, very desirable.

Lamarque, flowers very large, white with a sulphur yellow centre, plant of strong growth, requires a dry sheltered situation with a rich soil, and to be protected during the severe weather about Philadelphia.

I have omitted many others nearly equal to the above, such as Solfatare, Du Luxemburg, Phaloe, Boulogne, *Augusta*, &c. Of the latter I may say that I have not yet conversed with a judge or no judge of roses, in this vicinity, who pronounces it equal or nearly equal to the published description of it. I

paid a V. for my plant, and consider it fully paid. If I was now to make out a list to send to my Florist, I would say Cloth of Gold, Ophirie, Augusta, &c.

[To be continued.]

THE STRAWBERRY CONTROVERSY.

The communication of Mr. Meehan to the Pennsylvania Horticultural Society, seems to have excited the opposition of all those persons, and we believe they are the majority, who have settled the question to their own satisfaction, that the Strawberry is either pistillate or staminate, and unchangeably so. Mr. Meehan has produced his plants (of Hovey's Seedling), with staminate flowers. The Strawberry cultivators pronounce them (in the teeth of Mr. M's assertion that they are runners from pistillate plants,) to be not Hovey's, but another variety. Some others talk of the absurdity of plants being fruitful without fertilization, which, whether absurd or not, Mr. M. does not claim for his plants; we have seen nothing yet on that side in the way of argument, it is all assertion;—The Cinn. Hort. Society, formally pronounced it impossible, and with them there is no appeal from their “ipse dixit.”

In the last number of the Farm. Journal, we have a letter, which we suppose is from the eminent botanist of that region, which we copy, as suggesting many reasons why Mr. Meehan may be right.

The statement of MR. MEEHAN, in the April number of the Farm Journal, alleging that he has observed the sexual characters of the Strawberry flowers to be variously modified by culture, or different methods of treatment—has elicited some strong asseverations of *opinion*, in contradiction to that allegation of *fact*. One writer unhesitatingly declares the alleged change to be “*utterly impossible*,” and I understand that in the Queen City of the West, they have had a *public gathering*, to deliberate on the subject, which resulted in a *Pronunciamento* adverse to MR. MEEHAN'S statement,—his facts and observations being *rejected* by a clear majority of the voters present! The matter being thus *settled*, by preamble and resolution after the manner of political difficulties at a war-meeting, it may seem to be out of order, now, to offer any remarks on the controverted topic. Nevertheless, as this is reputed to be a Free Country, I should like to be indulged with the privilege of submitting a few suggestions,—if not in arrest of judgement, at least as a plea in mitigation of the sentence, against my friend MEEHAN. It is the remark of a vigorous and sagacious modern writer, that “no scientific question was ever yet settled dogmatically, nor ever will;” and I think the same may be especially predicated of questions of fact, in Natural History. I may here observe, that I was favored with the opportunity of examining one of MR. MEEHAN'S specimens,—in which there were certainly two scapes from the same root—one bearing a cyme of *pistillate* flowers (with minute rudiments of abortive stamens,) and the other a cyme of *perfect*, or hermaphrodite flowers: and whether the specimen was the progeny of a pistillate, a staminate, or a hermaphrodite plant, I should think the inference plausible, that the flowers on at least *one* of these two cymes, must have been a modification, or altered product, of the parent plant. It is this kind of change, in the character of the flowers, which I understand MR. MEEHAN to announce, as having occurred in plants under his

management. Now, in view of the countless modifications daily observable in the organs of plants—and especially in the *floral organs*—I can perceive no sufficient ground for declaring the changes, reported by MR. MEEHAN, to be “utterly impossible.” The modifications here referred to, are a very different thing from the alleged *transmutation of one kind to another*,—which is vulgarly supposed to take place in certain plants, just as the Alchemists formerly pretended was affected among the metals. They merely alter the texture, distort the forms, or affect the developements of organs; but do neither change nor annihilate those *essential characteristics*, by which the plant is rendered permanently distinct from every other genus and species. The floral organs of many plants are remarkably subject to modification, under the long-continued influences of soil, climate, and culture, or management. Some flowers are rendered *double*, as it is termed, by the expansion of stamens into petals; others become imperfect, and even neutral, by the abortion or blighting of the stamens, or pistils, or both. The *Strawberry* appears to be very liable to this kind of blight; and hence the much talked of sorts, among cultivators, of *pistillates*, and *staminates*,—though in all the pistillate flowers, which I have examined, there were *vestiges* more or less obvious, of abortive *stamens*, on the rim of the calyx. It also varies much, under culture, in some other features,—especially in the developement and character or quality of the *receptacle*, or what is commonly regarded as *fruit*: but no one, I believe, has yet seen a Strawberry plant transmuted into a *Cinquefoil*, though so nearly allied in habit. The organs of plants may be greatly disguised by the influences above mentioned; but still the essential distinguishing traits are preserved,—and there seems to be no insuperable obstacle to prevent a plant, with modified or abortive organs, from reverting, under a change of circumstances, to its original condition, and resuming its pristine form and character. The normal, or what may be called the *constitutional* character of the Strawberry-flower, is to be *perfect*—*i. e.* furnished with both stamens and pistils (possibly such may be the true normal structure of *all* flowers); and although many other plants, as the Strawberry, are found with *imperfect*, and even *neutral* flowers,—every Naturalist and careful observer knows, that there is often an obvious effort and tendency, in such flowers, to a more complete development: *i. e.* to become *perfect* and *regular*. We occasionally see *diandrous* flowers become *didynamous*,—and *didynamous* plants developing *regular petandrous* flowers; and it is not at all unusual to find the *staminate tassel* of the cultivated Maize (*Zea Mays*, *L.* a *monoicous* plant,) bearing *fertile flowers*, and exhibiting a very successful attempt at the production of an *Ear*—or *cluster of Ears*—of *Indian corn*. These instances, I trust (for it is needless to multiply them,) may suffice to show that there is nothing irrational, nor incredible, in MR. MEEHAN’S observations; and that it is rather strong phraseology, to declare such phenomena to be “utterly impossible.” The polemic writers on this Strawberry question, speak of the necessity of staminate plants among the pistillate, to produce, or perfect the *fruit*. No doubt, the pistils must be fertilized, in order to produce *seeds* that will vegetate. But what do those gentlemen understand, by the “*fruit*?” Do they mean the little single-seeded *akenes* or *nutlets*, which are sprinkled over the enlarged pulpy receptacle,—or do they refer to the *receptacle itself*, which in popular parlance is intended by the term “*fruit*?” If they have reference to the *real fruit*—the *nutlets* which contain the seed,—there is probably no question (as already intimated) about the necessity of staminate influence to produce perfect fruit. But I have a suspicion, that by the term “*fruit*,” they mean the delicious *receptacle* which bears the fruit, and if they mean to allege that the pistils must be fertilized by the stamens, *in order to produce that enlargement of the receptacle* which affords an *esculent substitute* for fruit,—then I have only to say, it is a *question of fact* which I have had no adequate opportunity to determine; and concerning which I, for one, should be happy to receive reliable information. To ascertain the point satisfactorily, would require very careful experiments and observations. Whether such have been made, I am not informed. I may remark, however, by way of *analogy*, that there are instances in which *pistils*, and even *receptacles*, are enlarged, where no staminate influence has been exerted. The conglomerate coalescent *pistils* of the Osage Orange (*Maclura*,) for example, attain to their full natural size (although the *seeds* are necessarily imperfect,) where no staminate plant is in the neighborhood; and, what affords a closer analogy, the including *receptacle* of the *pistillate Fig* is fully developed; when

entirely free from any staminate influence. Whether the *receptacle* of the *Strawberry* ever enlarges, without the pistils being fertilized, (as already stated,) is more than I can tell; but I feel well assured, that any competent authority, who may furnish the information, will make an acceptable contribution to physiological Botany.

W. D.

West Chester, June 6, 1853.

THE STRAWBERRY QUESTION.

MR. EDITOR:—Who among horticulturists has not heard of the “Strawberry Question”—has not wondered at the opposite opinions held by eminent cultivators, and perhaps like myself has been surprised that the spirit of inquiry has not been more generally diffused concerning a subject of so much importance. Is it because our instructors cannot come to a unanimous conclusion upon it, that we pupils in horticulture have been doubtful about expressing our opinions upon it, or have we never studied the subject for ourselves but left them in their wisdom to decide it for us? I was in this position waiting for their decision, but as it was not likely to be forthcoming, I resolved to study the subject for myself, have done so this season, and will give you my experience in connection with the “Strawberry Question.”

The points of difference between a staminate and pistillate strawberry flower are so marked that a very casual observer cannot fail to notice them at once, for in staminate varieties the flowers are lax, the sepals of the calyx appear alternately with the petals of the corolla, the stamens occupy the most prominent place in the flower, rising half their length above the central pistils, are very much swelled at the base, and very large compared with the other parts of the flower, anthers broadly heart shaped, large, and producing abundance of pollen. Pistils, loose, long, and having a barren appearance, which is soon proved by their withering away, and leaving the stamens masters of the field. The embryo receptacle when it does exist at all is very much flattened, and in the most of cases cannot be said to exist at all, the pistils being inserted in the thickened cup of the calyx. All flowers that have this appearance in these varieties, very soon wither away and leave this impression on the mind of the observer, that they were surely staminate varieties. But as all staminate varieties do not thus wither away, but many of them produce fruit, it may be asked are the flowers all the same in appearance. It is here where the peculiarities in this class are met with, for there are few but what produce some fruit, and as real staminate flowers could not produce this fruit, it would readily be seen that staminate varieties produce two sorts of flowers; the one where the organs are unequally balanced which produce no fruit, the other where the flowers are perfect and are fruit bearing. The appearance of the flower that will produce fruit is markedly different from the other—by the shortness of the stamens, the conical shaped receptacle, the pistils stiff and thickly set thereon. The flower is altogether more compact, and is as perfect in its organs as any of the Alpine varieties. The following sorts have proved staminate with me.

Cuthill’s Black Prince, Boston Pine, Kittley’s Goliath, Alice Maud, British Queen, Ross’s Phœnix, and Victoria.

Among pistillate varieties, the flowers are cup shaped, compact, with the sepals of the calyx scarcely perceptible between the petals of the corolla. Stamens seldom visible, and when seen at all they are very minute, not longer than the pistils at the base of receptacle, never showing signs of having fertilizing powers, as the anthers never expand; the whole remains perfectly abortive. Pistils very numerous, uniform, stout, longer than in perfect flowers, and have not such a feathery appearance as in staminate sorts. Receptacle large, conical, always coming to perfection, and the instances where pistillate flowers do not produce berries are very rare, in fact scarcely ever to be met with; at least my sorts have proved so; and it is reasonable to conclude that this has been occasioned by the sorts growing together. The following sorts have proved pistillate with me.

Bourbon Pine, Hudson, Swanson, Burr’s Pine, Hovey’s Seedling, Iowa, and Moyamensing.

The varieties enumerated, I have growing together (both pistillate and staminate) under what might be called the ordinary mode of cultivation in one place; and in another completely isolated from them, I have a collection of the same sorts which have stood some years longer, and are almost worn out. But this difference of situation, culture, &c. don't at all seem to have affected or in the least degree altered the organs of re-production in any variety; for to me it seems that the character of pistillate or staminate is as permanent and unchangeable as in any other deciduous plant.

A writer in one of the periodicals of the day, supposes it possible that the receptacle of the strawberry might enlarge, or even come to perfection without staminate influence having at all been required. That the experiment of planting a pistillate strawberry in a situation where staminate influence could not reach it and that plant produce no berry, is the fact; as was proved by a cultivator in this neighborhood: but it is to be hoped that the coming season will be taken advantage of by many of your correspondents, for making experiments that will place this question in such a clear and forcible manner before the public, so that all may be convinced, and those interested in the culture of this fruit may take advantage of, and turn to good account the information received.

F.

We hope that those of our friends who have made any observations, or have anything to say on this interesting subject will be kind enough to communicate them to us.—ED.

AMMONIA ON PLANTS.

The subject of the action of Ammonia on plants is exciting considerable attention in England. We copy below from the Gardener's Chronicle accounts of experiments, the first of which is being tried at the Horticultural Society's Garden.

M. Ville's mode of giving Ammonia to plants, with a view to increase their bulk and vigour, is being tried in the large stove in which one of his apparatuses has been placed. It consists of two clear glass bottles with long necks, furnished with tight-fitting corks, in each of which is inserted a small bent glass tube. These two tubes are joined together by means of an India-rubber connection, or small hose, thus forming a communication between the two bottles. In the cork of one of the bottles is an escape tube (also of glass), which is connected (by means of a small India-rubber hose), with other small glass pipes that are laid all along and across the bed, and through which the ammonia is intended to pass, in order that it may be the better diffused among the plants. When the bottles are put to work, one is charged with chalk, on which is poured sulphuric acid, and the other with unslacked lime, over which is poured a solution of ammonia. The result of this experiment will, of course, be published in due time. Its conduct has been entrusted to Mr. Spriggs, the young man in charge of the house, who is to note down its effects daily, and report the same to the Vice-Secretary. In another column will be found some further account of furnishing plants with more ammonia than they can get under ordinary circumstances.

By Mr. Deane, Vice President of the Pharmaceutical Society. Effects analogous to those produced by M. Ville (see last year's volume, p. 755), with ammoniated air on the leaves of growing plants, have been observed by me, as the results of applying solutions of ammoniacal salts to the roots. My attention was first effectively turned to the subject about eight or ten years since, when an extensive grower of Pelargoniums, Fuchsias, and Roses, applied to me for some remedy for the sickly condition of his stock; which, if left unchecked, would insure a very severe loss to him. On examining the plants they were found to be in a starving condition, the roots having filled the pots and exhausted the soil; consequently, the leaves had lost their healthy green colour, and become very pale, with a strong tinge of yellow; the lower leaves were quite yellow, spotted, and falling off. The natural remedy was obviously fresh potting, but as the plants were already in pots best adapted to answer the purposes of the grower, some other remedy had to be devised. I therefore made a very weak solution of sulphate and carbonate of ammonia, and therewith watered the roots of the plants once a day, in the evening; and to insure any observed results as to the effect of the ammonia, certain rows of the plants on the stage of the greenhouse were selected for the experiment. In a few days the effects of the ammonia were most marked and satisfactory. The leaves began to put on a very remarkable appearance, the course of the veins, or spiral vessels, becoming perfectly green, the colour commencing at the basal portion of the midrib, and thence spreading through all the reticulations, until the tissues were perfectly restored to their normal and healthy condition; and, in fact, the plants thus treated looked more vigorous than they had ever done before, being much darker colour and firmer in texture. The contrast between these plants and those which had received no ammonia left no doubt about the efficiency of the application. I forget the effects upon the flowering of the Pelargoniums, but there was certainly no deficiency of flowers on the Fuchsias and Roses; they were, moreover, finer and better coloured than usual. On a subsequent occasion a gentleman's

gardener applied to me in a similar dilemma; he had a house full of fancy Pelargoniums preparing for a flower-show, at which he expected to take the first prize. Just as the trusses of flower buds were emerging, and there was every prospect of a good bloom, the lower leaves of the plants began to turn yellow and spotted, and then to fall off, leaving the plants bare, where the foliage was considered an essential point of beauty. I examined the roots and found them nearly filling the pots, it was therefore evident there was not sufficient nutriment left in the pots to meet the extra demand made by the large number of flower-buds; the latter were, consequently, deriving their nourishment from the leaves—the natural storehouse of the food of plants during the growing season—and of course exhausted the lower leaves first. They were treated precisely as in the former instance, and with the same results; the lower leaves became healthy, and the flower-buds progressed favourably to maturity, being of good form and colour. The success of these experiments became known to other gardeners in the neighbourhood, some of whom were equally successful, while others did not derive that satisfaction from the use of the ammoniacal solution, either from not understanding the principle of its application, or from a desire to accomplish more than they were capable of, when it frequently happened the plants became too vigorous to flower well. There is no doubt but that M. Ville is correct in stating that the flowering is arrested if the application of ammonia is made at a certain period of the development of the flower-buds. Few plants if grown too vigorously will flower well, if at all. A certain check in their growth is absolutely necessary, and the summer's sun or winter's cold, under ordinary circumstances, effects this perfectly in this climate—the former by perfecting and condensing the elaborated sap, and the latter by arresting vegetation altogether. Too much moisture and shade cause those parts intended for flower-buds to be developed as leaves. In the Aloe tribe when the flower stem is thrown up, it is at the expense of the outer leaves, the elaborated juices of which it appropriates, the roots at this time not being in action, because it is towards the close of a long period of dryness. If when the flower-stem is beginning to rise, the roots are watered, all further development of the stem is arrested, the leaves only being developed. The same thing takes place with many bulbs whose period of flowering is not the same as that for leafing. Many Cape bulbs follow this law; for example, the Hæmanthus, the flowering of which is at the expense of some one or more of the outer coats. If these plants are watered at the wrong period, or if they have had not that proper rest which Nature designed they should have under the influence of a roasting sun, such as their native country affords, no flowers will be produced, but in their stead a vigorous development of leaves. It would appear, therefore, that the arrest of development of the flowers and fruits of the plants treated with ammonia, is not so much the result of any specific property possessed by this substance, as by its bringing about artificially those conditions which may occur naturally, or be produced by other means. Also, that the application of ammonia to plants may be attended by results varying according to the conditions under which it is applied, and the object it is desired to obtain. The following is the formula for the solution alluded to in the previous note by Mr. Deane:—Sulphate of ammonia, 7000 grains; sesquicarbonate ditto, 1000 grains; water, 80 fl. oz. Dissolve. Of this solution one fluid ounce to a gallon of water will make a solution, sufficiently strong for all ordinary purposes.—*Pharmaceutical Journal*.

CALENDAR OF OPERATIONS.

DISBUDDING.—This operation does not appear to be sufficiently recognised in its proper sense as distinguished from pruning and pinching. In performing these latter operations we remove a portion of the growing shoot, in the former case the young bud is removed as soon as it can be rubbed off. These operations are therefore quite distinct, and their distinction is of much importance. Most fruit cultivators are aware that trees suffer materially by suddenly depriving them of a large portion of foliage while in active growth, and expedients are resorted to in order to render the operation less injurious. In spring when the buds burst, attention should be directed to the quantity of young shoots desirable either for fruit or uniformity of growth; these being secured all others should immediately be *rubbed* off. As growth proceeds luxuriant shoots are *stopped* or their points pinched off, removing more or less of the shoot, according to the object to be attained. If the plant is very luxuriant, more leaves may be removed and if *every* shoot upon a tree is operated on in this manner it amounts to a severe check on its growth. In the case of young trees, or weakly ones, where a certain form is desired, the young growing point should be bruised without removing any of the elaborating foliage, securing density of habit without any perceptible check of growth. Suppose a shoot that has grown 12 or 14 inches to be pinched back one half of its length, the uppermost bud will burst again and the others remain comparatively dormant; but allowing the same shoot to have been checked in its longitudinal growth, by pinching or bruising its extreme point, it will be found that all the lower buds will be benefitted and several additional shoots produced. We have alluded to this subject more particularly at present as we are aware of having occasionally made use of the term *pinching* when *disbudding* would have been the more appropriate expression.

STRAWBERRIES.—Preparations should now be in progress for securing plants for early forcing next spring. Various methods are adopted to get strong plants, such as filling small pots with rich soil and plunging them in the strawberry patch, introducing a young plant on the pot, and removing into larger ones when rooted. Others again prepare full sized pots at once

and place them in this manner, securing the young plant in both cases with a small stone laid on the surface. When close attention is paid in watering, these plans are good. An equally successful and less troublesome method is to prepare a few square yards of ground in a somewhat sheltered situation, manuring it well and digging deeply, and filling it thickly with young plants. They should be partially shaded for a few days, and duly watered. In the course of three or four weeks they will lift with good balls of earth and are potted at once in fruiting pots. Pistilo-staminate or hermaphrodite varieties are found to produce better when forced early than pistillate sorts.

Root pruning of fruit trees is sometimes commendable, and the present is the proper season for its more immediate beneficial effects. It is mostly performed on young vigorous trees that show no disposition to fruit. By cutting away some of the strongest roots at this time the supply of sap will be lessened and the wood ripening process accelerated. No fruit need be expected from badly ripened wood. In nine cases out of ten the non-appearance of fruit on healthy trees arises from this cause. Mild autumn weather induces growth to a late period and sudden frosts arrive before the wood is sufficiently matured. This is more particularly noticeable on soils inclined to be wet, and here again we perceive the necessity of underground drains, in order to remove surplus water. It is an exceedingly erroneous idea that drains are worse than useless in localities where long droughts are of frequent occurrence. The truth is that draining enhances humidity during hot weather, since it enables the soil to exercise its absorbing properties to the fullest extent, there being more air in the soil water is retained in its pores, constituting a reservoir holding a lasting supply when other sources fail.

RENOVATING OLD TREES.—Most satisfactory results have been obtained from old and apparently worn out trees by changing the soil about their roots, or applying a top dressing of wood ashes, guano, salt and plaster in equal quantities, allowing about one bushel of the mixture to each tree. If nothing better is convenient, a heavy dressing of well made barn manure forked in about the roots will have a decided effect. The absorbing points of the roots extend a considerable distance from the trunk, therefore, the principal part of the top dressing should embrace a circumference at least equal to that of the branches. It might be considered unnecessary to mention

this very apparent fact, were it not usual to observe manure applied to the stem instead of the roots.

GRAPES UNDER GLASS will now be approaching maturity. Dryness both in soil and atmosphere favors this process, still they must not be allowed to suffer for want of moisture should the weather prove dry. The young growth may be stopped more rigidly as the ripening process proceeds. This will concentrate the sap and strengthen the buds for a future crop, if pinched back too severely these buds may start into growth, which must be guarded against.

OUT-DOOR GRAPES will require attention in thinning out lateral shoots and stopping others especially those on bearing shoots. Thinning out the berries is seldom practised on these, nor, indeed is it always necessary. Occasionally, however, the berries are so thickly placed that room is not afforded them to swell out and ripen properly. By thinning out a few of the most central and smallest berries, the fruit will ripen earlier and be much improved both in size and flavor.

S. B.

AUGUST.

FLOWER GARDEN AND PLEASURE GROUND.

This is the month many prefer to plant their evergreens, and it is perhaps as good a time as any. Take advantage of a “wet spell” for the operation. Be particularly careful that as many roots as possible are preserved. If this can be well attended to, *trees of any size* can be moved successfully. It becomes a question of power—of profit and loss. If any amount of power can be applied, and expense is no object, the largest trees will move as easily as small ones. In moving an evergreen 20 feet high, I commence to open my “trench” 10 feet from the base of the trunk, go down about two feet, and continue to undermine and lay bare the roots their full length, right up to the collar of the tree. I care nothing for “ball of earth.” If I have a good supply of hands at the job, I only care to keep them aside a little to prevent injury by the operations of the workmen. If I have but few men, I roll the long roots, as fast as they are exposed, in mats to keep them moist. For mechanical means and adaptations to take out the tree and convey it to its

destination, an intelligent workman is never at a loss. Trees taken up in this manner scarcely miss the change; and besides after they are transplanted they require no staking, as they are capable of withstanding the strongest wind through their long roots.

The broom should be well applied to lawns at this season of the year, even more sedulously than in spring. Many species of weed become so dwarfed by the summer's drought, that they flower and fruit below the reach of the scythe, and can only be kept down by the hard sweeping. I have seen some lawns almost ruined by these weeds, especially by one of the nettle tribe. (*Pilea pumila*.)—Conclusion in next number.

It is a pity that the rules adopted by Florists are so very rigid, that unless a flower equals a certain standard in shape and markings it must be rejected. Last week we had submitted to us by Mr. T. F. Croft, a beautiful seedling Verbena, a lilac with a broad stripe of white down each petal, quite distinct and very handsome, but the petals were too narrow, leaving a large space between each one. This must condemn it as a Florist's flower, but to all who want a handsome and distinct variety it will be desirable. He calls it his No. 3. It is in style of Iphigenie.

The article on Tile draining in the last number should have been credited to the Ohio Cultivator. We never intentionally copy articles without credit. We wish that Dr. Warder of the Western Horticultural Review were as conscientious.

ERRATA.—In the report of the Maryland Horticultural Society, the Winter Bon Chretien pears were shown by Mr. Fuss, not Mr. Feast as printed. The premium for strawberries was for “the best 4 distinct varieties, 1 quart each”.

*** END OF THE PROJECT GUTENBERG EBOOK THE FLORIST
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